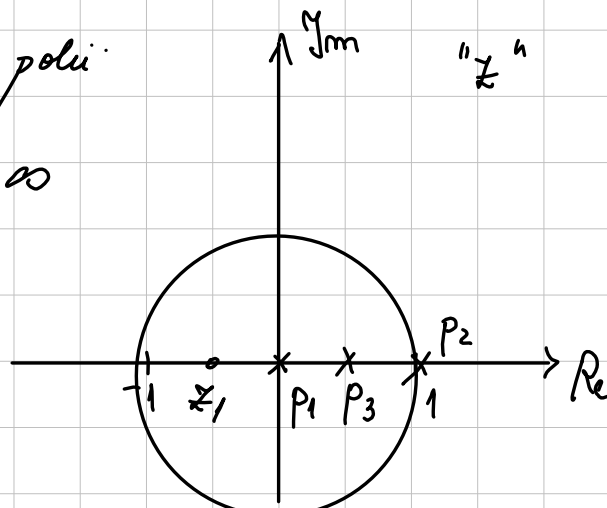


Recapitulare SAE

$$\underline{\text{ex 1}} \quad G(z) = \frac{0,4K(z+0,6)}{z(z-1)(z-0,36)} \quad \leftarrow \begin{array}{l} \text{zerouri} \\ \text{polii} \end{array}$$

Pasul 1: $m=3$ $(n-m)=2 \rightarrow \infty$
 $m=1$ $1k \rightarrow 0$



$$z_1 = -0,6$$

$$p_1 = 0; p_2 = 1; p_3 = 0,36$$

Pasul 2: $(-0,6, 0) \cup (0,36, 1)$

Pasul 3: $(n-m)=2$ asimp.

$$\varphi_k = \pm 180^\circ \frac{(2k+1)}{(n-m)} \Rightarrow \quad k=0 \quad \varphi_{k0} = \pm 180^\circ \frac{1}{2} = \pm 90^\circ$$

$$k=0,1$$

$$k=1$$

$$\varphi_{k1} = \pm 180^\circ \frac{3}{2} = 90 \cdot 3 = 270^\circ = -90^\circ$$

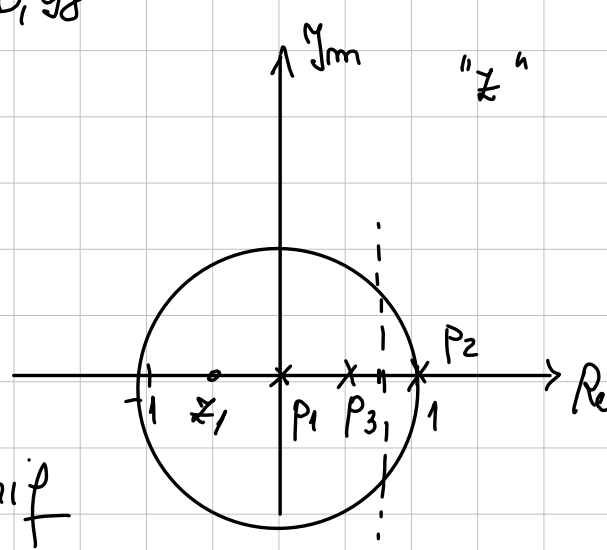
$$\overline{\sigma}_a = \frac{\sum p - \sum z}{n-m}$$

$$\overline{\sigma}_a = \frac{1,36 + 0,6}{2} = \frac{1,96}{2} = 0,98$$

Pasul 4: —

Pasul 5: pct. de ramif.

între 2 poluri
 există un pct. de ramif
 (între p_3 și p_2)



$$1 + G(z) = 0$$

$$1 + \frac{0,4K(z+0,6)}{z(z-1)(z-0,36)} = 0 \quad K = -\frac{z(z-1)(z-0,36)}{0,4(z+0,6)}$$

$$\frac{dk}{dz} = \frac{(z^3 - 1,36z^2 + 0,36z)' \cdot 0,4(z+0,6) - (z^3 - 1,36z^2 + 0,36z) \cdot 0,4}{(0,4)^2(z+0,6)^2}$$

$$\frac{dk}{dz} = \frac{-0,4(3z^2 - 2,72z + 0,36)/(z+0,6) + \overbrace{(0,4z^3 - 0,54z^2 + 0,14z)}^a}{0,16(z+0,6)^2}$$

$$\frac{dk}{dz} = \frac{(-1,2z^2 + 1,08z - 0,14)(z+0,6) + a}{0,16(z+0,6)^2}$$

$$= \frac{-0,66z^2 - 0,21z - 0,08 + 0,4z^3}{0,16(z+0,6)^2}$$

$$\frac{dk}{dz} = \frac{0,4z^3 - 0,66z^2 - 0,21z - 0,08}{0,16(z+0,6)^2} = 0$$

$$\Rightarrow z_1 \in \mathbb{R} \in \text{intervalul de la pct. 2}$$

$$z_2 \notin \text{pct. 2}$$

$$z_3$$

$$1 + G(z) = 0$$

$$z^3 - 1,36z^2 + 0,36z + 0,4Kz + 0,24 = 0$$

$$\left(\frac{n+1}{n-1}\right)^3 - 1,36 \left(\frac{n+1}{n-1}\right)^2 + 0,36 \frac{n+1}{n-1} + 0,4K \frac{n+1}{n-1} + 0,24K = 0$$

$$(n+1)^3 - 1,36(n+1)^2(n-1) + 0,36(n^2-1) + 0,4K(n+1)(n-1)^2 + 0,24K(n-1)^3 = 0$$

$$\begin{aligned} & n^3 + 3n^2 + 3n + 1 - 1,36(n^3 + n^2 + 3n - 1) + 0,36(n^2 - 1) + \\ & + 0,4K(n^3 - n^2 - n + 1) + 0,24K(n^3 - 3n^2 + 3n - 1) \\ & - 0,36 + 2,64n^2 - 1,08 + 2,36 + 0,36n^2 - 0,36 + 0,64Kn^3 - \\ & - 0,32Kn^2 + 0,32Kn + 0,16 = 0 \end{aligned}$$

$$-0,36n^3 + 3n^2 - 1,08n + 2 + 0,64n^3 - 0,32Kn^2 + 0,32Kn + 0,6K = 0$$

$$n^3(0,64K - 0,36) + n^2(3 - 0,32K) + n(-1,08 + 0,32K) + 0,16K + 2 = 0$$

Pasul 6:

$$G(z) = \frac{0,4K(z+0,6)}{z(z-1)(z-0,36)}$$

$$0,64K > 0,36$$

$$K > \frac{0,36}{0,64}$$

$$K > \frac{9}{16} = 0,56$$

$$3 > 0,32K$$

$$\frac{3}{0,32} > K$$

$$9,37 > K$$

$$0,32K > 1,08$$

$$K > \frac{1,08}{0,32}$$

$$K > 3,37$$

$$0,16K > 2$$

$$K > 12,5$$

$$K \in (3,37; 9,37)$$

$$n^3 \quad \left| \begin{array}{cc} 0,64K - 0,36 & -1,28 + 0,32K \end{array} \right.$$

$$n^2 \quad \left| \begin{array}{cc} 2 - 0,32K & 0,16K + 2 \end{array} \right.$$

$$n^1 \quad \left| \begin{array}{cc} \text{gm. 2} & 0 \end{array} \right.$$

$$n^0 \quad \left| \begin{array}{cc} 0,16K + 2 & 0 \end{array} \right.$$